9th Class • Punjab Board • PCTB New Syllabus

Chapter 7: Enzymes

Solved notes in book order — MCQs, short answers, detailed answers and key points.

Chapter 7 • Units 7.1 – 7.5 • Exercise solved in book order

What this chapter is about

• Metabolism is the sum of all chemical reactions in an organism. Catabolism breaks big molecules into smaller ones and releases energy; anabolism builds bigger molecules and uses energy.

• Enzymes are biological catalysts — mostly proteins — which speed up the reactions of the body without being consumed. Each enzyme has an active site which fits only its own substrate, so enzymes are highly specific.

• The chapter explains the lock and key and the induced fit models, the factors that affect enzyme activity (temperature, pH, substrate concentration), and the two types of enzyme inhibition.

• The exercise (10 MCQs, 10 short answers and 5 detailed answers) is solved here in book order.

A. Multiple Choice Questions

Tick the correct option. The reason after each answer tells you why the other options are wrong.

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1. Primarily, all enzymes are;

a) Nucleic acids
b) Proteins
c) Carbohydrates
d) Lipids

Answer: b) Proteins — Almost all enzymes are proteins (a few RNA molecules, called ribozymes, also act as enzymes). Enzymes are not carbohydrates or lipids.

2. Which best defines an enzyme?

a) A chemical that breaks down food
b) A hormone that regulates metabolism
c) A protein that speeds up reactions
d) A molecule that stores energy

Answer: c) A protein that speeds up reactions — An enzyme is a biological catalyst: it speeds up a chemical reaction without being used up. Breaking food is only one of its jobs, and it is not a hormone or an energy store.

3. What can happen if an enzyme is exposed to temperature that is higher than its optimal temperature?

a) Enzyme activity rate will increase
b) Enzyme's shape will change, potentially reducing its activity
c) Enzyme will speed up the reaction and remain stable
d) Enzyme will become a substrate itself

Answer: b) Enzyme's shape will change, potentially reducing its activity — Above the optimum temperature the weak bonds of the enzyme break, its globular

shape and active site are lost (denaturation) and its activity decreases.

4. Enzymes are specific in their action because;

a) Their active sites fit specific substrates
b) They are always proteins
c) They are consumed in reactions
d) They work only at high temperatures

Answer: a) Their active sites fit specific substrates — The active site has a fixed shape which only a particular substrate can fit, so one enzyme catalyses only one reaction. Enzymes are not consumed in the reaction.

5. Prosthetic groups are;

a) Required by all enzymes
b) Proteins in nature
c) Loosely attached with enzymes
d) Tightly bound to enzyme

Answer: d) Tightly bound to enzyme — Prosthetic groups are the organic cofactors which are tightly bound with the enzyme, e.g. vitamins, biotin and the haem group. Loosely bound organic cofactors are called coenzymes.

6. How does increasing temperature affect enzyme activity?

a) Increases activity to a point
b) Always decreases activity
c) Makes enzymes non-functional
d) No effect on enzyme

Answer: a) Increases activity to a point — Up to the optimum temperature the rate increases, because the molecules move faster; after the optimum the enzyme is denatured and the rate falls.

7. How does competitive inhibitor affect enzyme action?

a) Attaches with the substrate
b) Changes enzyme shape
c) Attaches and blocks the active site
d) Blocks the cofactors

Answer: c) Attaches and blocks the active site — A competitive inhibitor resembles the substrate; it competes with the substrate and attaches to the active site, so the substrate cannot bind.

8. An enzyme works best at a pH of 7.4. It is placed in an acidic solution with a pH of 4.0. How will this affect the enzyme?

a) The active site will be modified, reducing substrate binding
b) The enzyme will catalyse reactions faster due to increased H ions
c) The enzyme will gain additional active sites
d) The substrate will become inactive in an acidic environment

Answer: a) The active site will be modified, reducing substrate binding — A pH far from the optimum changes the ionisation (charge) of the amino acids of the active site, so the active site is changed and the enzyme cannot bind the substrate properly.

9. What is TRUE according to the induced fit model of enzyme action?

a) Enzyme's active site changes shape to bind the substrate
b) Substrate changes shape to bind to active site

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c) No shape changes occur in active site or substrate

d) Substrate attaches the enzyme at a site other that active site

Answer: a) Enzyme's active site changes shape to bind the substrate — In the induced fit model (Koshland, 1958) the active site is not rigid; when the substrate comes near, the active site is reshaped (induced) to fit it closely.

10. What is TRUE about the optimum pH values of the following enzymes of digestive system?

a) Pepsin works at low pH while trypsin works at high pH

b) Both work at high pH

c) Both work at low pH

d) Pepsin works at high pH while trypsin works at low pH

Answer: a) Pepsin works at low pH while trypsin works at high pH — Pepsin of the stomach works best in acidic medium (pH 1.5 to 2.0) and trypsin of the small intestine works best in alkaline medium (pH 7.0 to 8.0).

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B. Write Short Answers

1. Define metabolism. Differentiate between catabolism and anabolism.

Metabolism is the sum of all the chemical reactions that occur within an organism so as to sustain life. It has two sub-sets: catabolism and anabolism.

Catabolism	Anabolism
It is the breakdown of complex molecules into simpler ones, releasing energy.	It is the building up of complex molecules from simpler ones, using energy.
Energy is set free and stored in ATP.	Energy (ATP) is consumed.
Example: cellular respiration, in which glucose is broken into CO2 and H2O and 40.0 kcal of energy is set free.	Example: photosynthesis, in which CO2 and H2O are joined to make glucose; also protein synthesis from amino acids.
It is a destructive process (digestion and respiration).	It is a constructive process (growth and storage).

2. Which type of metabolism demands input of energy? Give an example.

Answer: Anabolism

Anabolism builds up complex molecules from simpler ones, so it consumes energy (ATP). Example: photosynthesis — carbon dioxide and water are joined to make glucose using sunlight, and oxygen is produced as a by-product. Another example is protein synthesis, in which amino acids are joined to make proteins.

3. Define an enzyme. What is its role in metabolism?

Definition: an enzyme is a biological catalyst which speeds up a chemical reaction in living organisms without being consumed in the process. Enzymes are primarily proteins and are highly specific to their substrates — they can speed up millions of times faster than uncatalysed reactions.

Role in metabolism: the reactions of metabolism (both catabolism and anabolism) would occur at an extremely slow speed at the body temperature without enzymes. Enzymes make them fast enough for life: digestive enzymes break down food, enzymes of respiration release energy from glucose, and enzymes of anabolism build proteins, fats and other molecules. So every step of metabolism is controlled by a specific enzyme.

4. What is the active site of enzyme? State its importance in enzyme specificity.

Active site: an enzyme has one or more pockets or clefts on its surface which are called active sites. The active sites are directly involved in catalysis and only a small part of the enzyme molecule (the catalytic region) forms them; the rest of the enzyme keeps this region in the right shape.

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Importance in specificity: the active site has a fixed shape which fits only a particular substrate, just as a lock fits only one key. Therefore a substrate can bind with an enzyme only when its shape is complementary to the active site. That is why amylase acts only on starch and not on proteins, and one enzyme catalyses only one kind of reaction.

5. Provide an example of a specific enzyme-substrate pair.

Enzyme: amylase. **Substrate:** starch.

The enzyme amylase specifically catalyses the breakdown of starch into simple sugars (maltose and then glucose). It does not act on proteins or fats. Other examples: pepsin acts on proteins, lipase on lipids (fats) and catalase on hydrogen peroxide.

6. What is the effect of substrate concentration on enzyme activity?

An increase in substrate concentration increases the rate of reaction, because more substrate molecules are available to occupy the active sites of the enzymes. At a high substrate concentration, however, the rate stops increasing: at that point all the active sites of the enzyme molecules are occupied. This state is called **saturation** of the active sites, and after it the further addition of substrate does not increase the rate — the reaction rate becomes constant.

7. Provide two examples of enzymes that operate optimally at specific pH.

1. Pepsin: the protein-digesting enzyme of the stomach works in acidic medium, its optimum pH being 1.5 to 2.0.

2. Trypsin: the protein-digesting enzyme of the small intestine works in alkaline medium, its optimum pH being 7.0 to 8.0.

(Amylase of the mouth, with an optimum pH of about 6.8, is another example.) This shows that every enzyme has its own optimum pH, and away from it the enzyme works slowly or not at all.

8. What do you mean by optimum temperature and pH?

Optimum temperature: each enzyme works at maximum rate at a particular temperature, which is called its optimum (optimum temperature). For the enzymes of the human body it is about 37 °C (37.5 °C). Below it the rate falls, because there is less kinetic energy and fewer collisions; above it the enzyme is denatured and stops working.

Optimum pH: each enzyme works at its maximum rate at a specific pH of the medium in which it works; this pH is called the optimum pH. For example, pepsin works best at pH 1.5 to 2.0 (acidic) and trypsin at pH 7.0 to 8.0 (alkaline). A change in pH affects the ionisation of the amino acids of the active site and reduces the enzyme activity.

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9. Which type of enzyme inhibitors inhibit the enzymes without attaching to the active site?

Answer: Non-competitive inhibitors

Non-competitive inhibitors do not resemble the substrate and do not attach to the active site. They attach to some other site of the enzyme (the allosteric site), which changes the overall shape of the enzyme and so changes the shape of the active site as well. The substrate then cannot fit and the enzyme is inhibited. Examples: heavy metals (such as mercury) and certain drugs used in cancer therapy.

10. Differentiate between competitive and non-competitive inhibition.

Competitive inhibition	Non-competitive inhibition
The inhibitor resembles the substrate.	The inhibitor does not resemble the substrate.
It attaches to the active site of the enzyme.	It attaches to some other site of the enzyme.
It competes with the substrate for the same active site.	It changes the shape of the active site.
Increase in substrate concentration can reduce its effect.	Increase in substrate concentration cannot reduce its effect.
Examples: antibiotics (they compete with the substrate of bacterial enzymes), antimetabolites.	Examples: heavy metals such as mercury, cyanide and some drugs used in cancer therapy.

Table: Types of enzyme inhibition

Remember the difference

• Prosthetic group: organic cofactor tightly bound to the enzyme (e.g. vitamins, haem group).

• Coenzyme: organic cofactor loosely bound to the enzyme (e.g. NAD, NADP, vitamins).

• Cofactor: a non-protein molecule or ion needed by an enzyme to be fully active — inorganic (metal ions) or organic.

C. Write Answers in Detail

1. Describe the characteristics of enzymes.

Definition: enzymes are biological catalysts that speed up chemical reactions in living organisms without being consumed in the process. They are primarily proteins and are highly specific to their substrates. Their characteristics are:

1. Chemical nature: enzymes are predominantly proteins. They typically contain about 100 to 1,000 amino acids. Some RNA molecules, called ribozymes, also act as enzymes; ribozymes are mainly found in ribosomes and in some viruses, and they are also found in specific viruses and bacteria.

2. Globular structure: the enzyme possesses a three-dimensional globular structure, which allows them to form active sites that can bind specifically to substrates.

3. Specificity: enzymes are highly specific to the reactions they catalyse and also to the substrate. For example, the enzyme amylase specifically catalyses the breakdown of starch into simple sugars.

4. Speed up the reaction: enzymes increase the rate of reaction by lowering the activation energy, so the reaction becomes millions of times faster than the uncatalysed reaction.

5. Not consumed in the reaction: an enzyme is not changed or used up; after the reaction it can be used again and again.

6. Needed in very small amount: one molecule of an enzyme can catalyse the conversion of a huge number of substrate molecules, so only a small quantity is needed.

7. Intracellular and extracellular enzymes: enzymes are classified on the basis of their location — intracellular enzymes operate within cells (e.g. enzymes of cellular respiration) and extracellular enzymes are secreted outside the cells to catalyse reactions e.g. enzymes working in stomach for the digestion of food.

8. Need of cofactors: many enzymes require additional non-protein molecules to be fully active; such non-protein molecules are called cofactors.

9. Effect of temperature and pH: each enzyme works at its maximum rate at an optimum temperature (about 37 °C for human enzymes) and an optimum pH; extremes denature the enzyme.

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2. Describe how temperature extremes can inhibit enzyme activity and lead to enzyme denaturation.

Effect of low temperature: at low temperature the molecules of the enzyme and the substrate have little kinetic energy, so they collide rarely. The rate of reaction is therefore very slow. Low temperature does not usually destroy the enzyme; when the temperature rises again, the enzyme becomes active again.

Increase up to the optimum: as the temperature rises, the molecules move faster and collide more often, so the rate of reaction increases. Every enzyme has a maximum rate at a particular temperature called its optimum temperature, which is

about 37.5 °C for the enzymes of the human body.

Effect above the optimum (denaturation): when the temperature is raised well above the optimum, the heat energy breaks the weak (hydrogen and ionic) bonds that hold the globular protein of the enzyme in its special three-dimensional shape. As a result the enzyme molecule unfolds, and the specific shape of its active site is lost. The substrate can no longer fit and the enzyme cannot catalyse the reaction. This loss of the enzyme's shape and activity is called **denaturation**, and it is usually permanent — cooling the enzyme does not bring back its activity. That is why a high fever is dangerous and why food is cooked at high temperature to destroy the enzymes of microorganisms.

Temperature	Effect on enzyme
Very low (near 0 °C)	Activity very slow; enzyme is not destroyed (temporary inactivation)
Increasing up to optimum	Activity increases; more collisions between enzyme and substrate
At optimum (about 37 °C in humans)	Maximum activity
Above optimum	Shape of active site changes; activity decreases
High (e.g. 60–80 °C)	Enzyme is denatured permanently; activity is lost

3. How does pH affect enzyme activity?

1. Every enzyme has an optimum pH: each enzyme works at maximum activity at a specific pH called its optimum pH. For pepsin of the stomach it is pH 1.5 to 2.0 and for trypsin of the small intestine it is pH 7.0 to 8.0; amylase of the mouth works best at about pH 6.8.

2. Change in the charge of the active site: a change in pH changes the concentration of H+ ions, which changes the ionisation (charge) of the amino acids present at the active site. The active site is therefore modified and the substrate can no longer bind properly, so the enzyme activity decreases.

3. Change in the shape of the enzyme: a strong change in pH also disturbs the bonds that hold the shape of the enzyme protein; the enzyme is denatured, just as by high temperature.

4. Optimal pH fits the place of work: because of optimum pH, the digestive enzymes work in the correct order — pepsin works in the acidic stomach, and when the food reaches the alkaline small intestine, pepsin stops and trypsin starts its work.

5. The activity-pH graph: if the activity of an enzyme is plotted against pH, a bell-shaped curve is obtained; the peak of the curve shows the optimum pH, and on both sides of the peak the activity falls.

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4. Describe the factors that affect the activity of enzymes.

Enzymes are sensitive to their environment; the activity of an enzyme is affected by the following factors.

1. Temperature: each enzyme works at maximum rate at a specific temperature called its optimum temperature (about 37.5 °C for human enzymes). When the temperature rises to a certain limit, the heat adds to the movement of the molecules, so the rate of enzyme activity increases. But when the temperature is raised above the optimum, the heat breaks the bonds of the enzyme molecule, its globular shape is lost, the enzyme is denatured and its activity decreases; at high temperature it becomes non-functional.

2. pH: enzymes work at their optimum pH (pepsin 1.5–2.0 and trypsin 7.0–8.0). A change in pH affects the ionisation of the amino acids of the active site, which slows down the enzyme activity or blocks it completely; a strong change denatures the enzyme.

3. Substrate concentration: an increase in substrate concentration increases the rate of reaction up to a point, because more substrate molecules occupy the active sites. After the saturation of the active sites, no further increase occurs.

4. Enzyme concentration: if the substrate is present in plenty, an increase in the concentration of the enzyme increases the rate of reaction, because more active sites become available.

5. Inhibitors: certain substances called inhibitors bind to the enzyme and decrease or destroy its activity. Competitive inhibitors attach to the active site, while non-competitive inhibitors attach at another site and change the shape of the enzyme.

6. Cofactors and coenzymes: many enzymes need non-protein helpers (metal ions, vitamins such as NAD and NADP) to be fully active; without them the enzyme cannot work.

7. Presence of other substances: the products of the reaction, salts and other chemicals present in the medium can also affect the enzyme by changing the pH or by blocking the active site.

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5. Compare the lock-and-key and induced fit models of enzyme action.

1. Lock and Key Model of Enzyme Action:

This model was proposed by a German chemist, **Emil Fischer, in 1894**. According to it, the active site of an enzyme has a fixed structure. The substrate molecule fits precisely into it to form an **enzyme-substrate complex**. The enzyme then catalyses the reaction and the substrate is transformed into the products, which are then released from the enzyme. Here the enzyme is like a lock and the substrate like the key that fits it exactly.

2. Induced Fit Model:

This model was proposed by an American biologist, **Daniel Koshland, in 1958**. According to it, the active site of an enzyme is not rigid. When the substrate interacts with the enzyme, the active site is reshaped (induced) to perform its function — the enzyme wraps itself closely round the substrate, just as a hand

adjusts its grip on a glove. The enzyme-substrate complex is formed, the reaction takes place and the products are released; the enzyme then returns to its original shape.

Lock and Key model	Induced fit model
Proposed by Emil Fischer (1894).	Proposed by Daniel Koshland (1958).
The active site has a fixed, rigid shape.	The active site is flexible (not rigid).
The substrate fits exactly into the active site, like a key in a lock.	The active site changes (is reshaped) to fit the substrate.
It explains specificity but not why one enzyme can act on slightly different substrates.	It explains specificity as well as the flexibility of enzyme action and is accepted today.
The shape of the enzyme stays the same during the reaction.	The shape of the enzyme changes during the reaction and returns to normal afterwards.

Conclusion: both models agree that an enzyme-substrate complex is formed at the active site and that the enzyme is not consumed. The induced fit model is better because it explains the observed flexibility of enzymes.

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Points to Remember

Metabolism is the sum of all the biochemical reactions occurring in living organisms. It has two parts: **catabolism** (large molecules are broken into smaller ones, energy is released) and **anabolism** (small molecules combine to form large molecules, energy is used).

Enzymes are proteins that catalyse (speed up) biochemical reactions in living organisms. They are highly specific to the reaction and also to the substrate.

Intracellular enzymes operate within cells (e.g. enzymes of cellular respiration); **extracellular enzymes** are secreted outside the cells (e.g. enzymes working in the stomach for digestion).

Cofactors are non-protein molecules required by enzymes to be fully active. Inorganic cofactors include metal ions (iron, magnesium); organic cofactors are of two types: prosthetic groups (tightly bound, e.g. vitamins, biotin, haem group) and coenzymes (loosely bound, e.g. NAD, NADP, many vitamins).

Active site: the catalytic region of the enzyme molecule where the substrate binds. Its shape is specific for its substrate.

Lock and Key model (Emil Fischer, 1894): the active site has a fixed structure and the substrate fits into it exactly.

Induced fit model (Daniel Koshland, 1958): the active site is not rigid; it is reshaped when the substrate interacts, so as to perform its function.

Factors affecting enzyme activity: temperature (optimum for human enzymes about 37 °C; high temperature denatures the enzyme), pH (pepsin 1.5–2.0, trypsin 7.0–8.0), substrate concentration (saturation of active sites) and enzyme concentration.

Inhibitors are substances that interfere with and block an enzyme's activity. **Competitive inhibitors** resemble the substrate and compete with it for the active site (e.g. antibiotics). **Non-competitive inhibitors** bind to a site other than the active site and change its shape (e.g. heavy metals such as mercury, cyanide, drugs used in cancer therapy).

Uses of enzymes in industry: food industry (bread, cheese, wine, fruit juice), paper industry, biological detergents (proteases and lipases in washing powders), fermentation industry (yeast enzymes) and medicine (diagnosis and treatment).

Term	One-line meaning
Metabolism	Sum of all chemical reactions in an organism
Catabolism	Breakdown of complex molecules, energy released
Anabolism	Building of complex molecules, energy used
Enzyme	Protein catalyst of biochemical reactions
Substrate	The molecule on which an enzyme acts
Active site	Specific pocket of the enzyme where the substrate binds

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Term	One-line meaning
Optimum temperature/pH	The temperature/pH at which the enzyme works fastest
Denaturation	Loss of the enzyme's shape and activity by heat or extreme pH
Cofactor	Non-protein helper needed by an enzyme
Inhibitor	Substance that reduces or destroys enzyme activity

Table: Important terms of the chapter

How to attempt these questions in the paper

• MCQ: write only the correct option (a, b, c or d).

• Short question: 2 to 4 sentences; give the definition and one example (pepsin, amylase).

• Difference question: always draw a two-column table with headings.

• Detailed answer: definition, then numbered points, and draw the graphs of temperature and pH where asked. The names and years of the two models (Fischer 1894, Koshland 1958) carry marks.

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